

# Tharun Sridhar

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## SUMMARY

AI-focused Computer Science graduate with hands-on experience building end-to-end machine learning systems and FastAPI backends. Skilled in deep learning, computer vision, LLM integration, REST API design, JWT authentication, relational data modeling, and database migrations, with projects spanning medical imaging, inventory management, media platforms, and malware classification.

## TECHNICAL SKILLS

**Languages:** Python, Java, SQL

**Backend:** FastAPI, REST API Design, JWT Authentication, SQLAlchemy, Alembic, Asynchronous Programming, Postman

**AI / ML:** Machine Learning, Deep Learning, PyTorch, TensorFlow, Scikit-learn, EfficientNetV2, LLM Integration, Feature Engineering, Model Evaluation, OpenCV, Pandas, NumPy

**Databases:** MySQL, SQLite, Relational Database Design

**Frontend:** React.js, HTML, CSS

**Cloud & DevOps:** AWS, Google Cloud Platform, Git, GitHub

## EDUCATION

**Vellore Institute of Technology**

*Bachelor of Technology in Computer Science*

Vellore, India

*May 2026*

## PROJECTS

**NeuroScan AI** | *Python, FastAPI, TensorFlow, OpenCV, Groq LLM*

[GitHub](#)

- Built a FastAPI-served brain tumor MRI analysis pipeline combining a 4-model classification ensemble with an Attention U-Net for tumor segmentation, achieving a Dice score of ~0.88.
- Designed an adaptive lesion-aware fusion layer and a Diagnostic Reliability Index (DRI) that cross-validates Grad-CAM attention against the segmentation mask, gating predictions into Accepted / Caution / Specialist-Review tiers instead of returning a single confidence score.
- Integrated Groq LLM report generation and PDF export through FastAPI endpoints, with 4 pytest suites covering reliability gating, fusion, quality checks, and segmentation output.

**Inventra – Inventory Management System** | *FastAPI, SQLAlchemy 2.0, Alembic, JWT, SQLite*

[GitHub](#)

- Designed and built a role-based inventory backend (Admin / Manager / Employee) with JWT authentication using access and revocable, database-stored refresh tokens, managed through Alembic migrations.
- Implemented atomic, idempotent transaction flows for purchase-order receiving and sales-order completion, ensuring every stock-affecting action writes exactly one immutable inventory transaction record with no update or delete path.
- Built the full REST API surface – 51 endpoints across 13 routers (auth, users, products, categories, suppliers, purchases, sales, returns, damage, transactions, reports, notifications, dashboard) – with search, filtering, sorting, and pagination on core resources.

**PhotoShare API** | *FastAPI, Async SQLAlchemy, JWT, ImageKit*

[GitHub](#)

- Built a FastAPI backend for photo/video sharing with JWT authentication via fastapi-users, covering registration, login, email verification, and forgot/reset-password flows.
- Streamed multipart file uploads to a temp file and pushed them to the ImageKit CDN, storing post metadata with UUID-keyed async SQLAlchemy models.
- Implemented ownership-based authorization returning 403 on delete attempts by non-owners, and served the REST API and static frontend from a single FastAPI process to eliminate CORS overhead.

**Malware Family Classification with Deep Learning** | *Python, TensorFlow, EfficientNetV2, NumPy*

[GitHub](#)

- Built a feature engineering pipeline converting 13,747 PE executable samples (8,388 train / 1,480 validation / 3,879 test) across 31 malware families into grayscale image tensors, then fine-tuned EfficientNetV2-S with class-balancing to reach ~95% test accuracy, a macro F1-score of ~0.96, and a weighted F1-score of ~0.95.
- Applied statistical evaluation across confusion matrices and per-class precision/recall to validate consistent performance across all 31 malware families, not just aggregate accuracy.

## CERTIFICATIONS

**AWS Certified Solutions Architect – Associate (SAA-C03) Specialization**

Packt

**Model Context Protocol: Introduction & Advanced Topics**

Anthropic

**Claude Code in Action**

Anthropic